# MATH 111 Calculus 1, NUU, FALL 2026

Dr. Salnikova Chekeres **Tutorial 7**

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*Inverse trigonometric and hyperbolic functions (Sections 6.6–6.7, Lecture 14); maximum and minimum values (Section 3.1, Lecture 15); the Mean Value Theorem and the shape of a graph (Sections 3.2–3.3, Lecture 16).*

**Key facts to have at hand (Lectures 14–16)**

- **Inverse trigonometric functions — restrict first, then invert.** $\sin^{-1} x = y \iff \sin y = x$ and $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$ (domain $[-1,1]$); $\cos^{-1} x = y \iff \cos y = x$ and $0 \le y \le \pi$ (domain $[-1,1]$); $\tan^{-1} x = y \iff \tan y = x$ and $-\frac{\pi}{2} < y < \frac{\pi}{2}$ (domain $\mathbb{R}$). They are *not* $1/\sin x$, $1/\cos x$, $1/\tan x$. *Undoing:* $\sin^{-1}(\sin x) = x$ only for $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$; $\cos^{-1}(\cos x) = x$ only on $[0,\pi]$; $\tan^{-1}(\tan x) = x$ only on $(-\frac{\pi}{2}, \frac{\pi}{2})$. $\lim_{x\to\pm\infty} \tan^{-1} x = \pm\frac{\pi}{2}$.
- **Their derivatives:** $\left(\sin^{-1} x\right)' = \dfrac{1}{\sqrt{1-x^2}}$, $\left(\cos^{-1} x\right)' = -\dfrac{1}{\sqrt{1-x^2}}$ $(-1 < x < 1$; vertical tangents at $\pm 1)$, $\left(\tan^{-1} x\right)' = \dfrac{1}{1+x^2}$ (every $x$); with the chain rule, $u'$ on top and $u$ inside.
- **Hyperbolic functions:** $\sinh x = \dfrac{e^x - e^{-x}}{2}$, $\cosh x = \dfrac{e^x + e^{-x}}{2}$, $\tanh x = \dfrac{\sinh x}{\cosh x}$; $\operatorname{sech} x = \dfrac{1}{\cosh x}$, $\operatorname{csch} x = \dfrac{1}{\sinh x}$, $\coth x = \dfrac{\cosh x}{\sinh x}$. $\cosh x \ge 1$; sinh is odd, cosh even; $-1 < \tanh x < 1$. *Identity:* $\cosh^2 x - \sinh^2 x = 1$. *Derivatives:* $(\sinh x)' = \cosh x$, $(\cosh x)' = \sinh x$ — *no minus sign*; $(\tanh x)' = \operatorname{sech}^2 x = 1 - \tanh^2 x$.
- **Extreme values.** $f(c)$ is the *absolute maximum* on $D$ if $f(c) \ge f(x)$ for all $x$ in $D$; a *local maximum* if $f(c) \ge f(x)$ for $x$ near $c$ (on an open interval around $c$ — so never at an endpoint); minima likewise. *Extreme Value Theorem:* continuous on a closed interval $[a,b] \Rightarrow$ an absolute maximum and minimum exist. *Fermat:* a local maximum or minimum at $c$, and $f'(c)$ exists $\Rightarrow f'(c) = 0$; the converse is false ($x^3$), and an extreme value may sit where $f'$ does not exist ($|x|$). *Critical point:* $c$ in the domain with $f'(c) = 0$ or $f'(c)$ not existing — candidates, not answers.
- **The Closed Interval Method** ($f$ continuous on $[a,b]$): the values of $f$ at the critical points in $(a,b)$ and at $a$ and $b$; the largest is the absolute maximum, the smallest the absolute minimum.
- **Rolle** ($f$ continuous on $[a,b]$, differentiable on $(a,b)$, $f(a) = f(b)$): $f'(c) = 0$ for some $c$ in $(a,b)$. **The Mean Value Theorem** ($f$ continuous on $[a,b]$, differentiable on $(a,b)$): $f(b) - f(a) = f'(c)(b-a)$ for some $c$ in $(a,b)$ — a tangent parallel to the secant. *Consequences:* $f' = 0$ on an interval $\Rightarrow f$ constant there; $f' = g'$ on an interval $\Rightarrow f = g + C$.
- **The shape of a graph.** *I/D Test:* $f' > 0$ on an interval $\Rightarrow f$ increasing there; $f' < 0 \Rightarrow$ decreasing. *First Derivative Test* ($c$ a critical point of a continuous $f$): $f'$ changes $+ \to -$ at $c$: local maximum; $- \to +$: local minimum; no change: neither. *Concavity:* the graph above its tangents = concave upward; below = concave downward. *Concavity Test:* $f'' > 0 \Rightarrow$ CU, $f'' < 0 \Rightarrow$ CD. An *inflection point* is where $f$ is continuous and the concavity changes — found where $f''$ changes sign. *Second Derivative Test* ($f''$ continuous near $c$, $f'(c) = 0$): $f''(c) > 0 \Rightarrow$ local minimum; $f''(c) < 0 \Rightarrow$ local maximum; $f''(c) = 0 \Rightarrow$ *no information.*

## Part I · Inverse trigonometric and hyperbolic functions; extreme values (6.6–6.7, 3.1) · 50 minutes

**Warm-up I · True or false? (justify with one sentence or one counterexample)**

1. $\sin^{-1}\left(\sin \dfrac{5\pi}{6}\right) = \dfrac{5\pi}{6}$.
2. $\cos^{-1}(\cos x) = x$ for every $x$ in $[0,\pi]$.
3. $\tan^{-1} x = \dfrac{1}{\tan x}$.
4. $\sin^{-1} x$ is defined for every real number $x$.
5. $\lim\limits_{x\to-\infty} \tan^{-1} x = -\dfrac{\pi}{2}$.
6. $\dfrac{d}{dx} \cos^{-1} x = \dfrac{1}{\sqrt{1-x^2}}$ for $-1 < x < 1$.
7. $\cosh x \ge 1$ for every real $x$.

8. $\dfrac{d}{dx}\cosh x = -\sinh x$.
9. If $f$ is continuous on the closed interval $[a,b]$, then $f$ has an absolute maximum value on $[a,b]$.
10. If $f$ has a local maximum at $c$, then $f'(c) = 0$.

**Problem 1** *Inverse trigonometric functions: values, the quadrant, simplification.*

(a) Find the exact values: $\sin^{-1}\!\left(-\dfrac{\sqrt{3}}{2}\right)$, $\cos^{-1}\!\left(-\dfrac{\sqrt{2}}{2}\right)$, $\tan^{-1}\!\left(-\sqrt{3}\right)$, $\sin^{-1}\!\left(\sin\dfrac{3\pi}{4}\right)$.

(b) Find the exact values of $\tan\!\left(\cos^{-1}\dfrac{7}{25}\right)$ and $\cos\!\left(\tan^{-1}(-2)\right)$. Draw a right triangle for each — and decide the sign from the *range* of the inverse function before you write the answer.

(c) Simplify $\sin\!\left(2\tan^{-1}x\right)$ for every real $x$, and $\cos\!\left(2\sin^{-1}x\right)$ for $-1 \le x \le 1$.

**Problem 2** *Derivatives with inverse trigonometric functions.*

Differentiate. For every $\sin^{-1}(\dots)$, $\cos^{-1}(\dots)$ or $\tan^{-1}(\dots)$ say what the *inside* is before you write anything.

(a) $y = \sin^{-1}(5x)$

(b) $y = x^2\tan^{-1}(4x)$

(c) $y = \cos^{-1}\!\left(e^{-x}\right)$, $x > 0$

(d) $y = \tan^{-1}\dfrac{x}{3}$

(e) $y = \sin^{-1}x + x\sqrt{1-x^2}$

(f) The tangent line to $y = \cos^{-1}\dfrac{x}{3}$ at $x = \dfrac{3}{2}$.

**Problem 3** *Hyperbolic functions: values and identities (homework).*

(a) Find the exact values of $\cosh\!\left(\ln\frac{5}{2}\right)$, $\sinh\!\left(\ln\frac{5}{2}\right)$ and $\tanh\!\left(\ln\frac{5}{2}\right)$; then check that $\cosh^2\!\left(\ln\frac{5}{2}\right) - \sinh^2\!\left(\ln\frac{5}{2}\right) = 1$.

(b) Prove, from the definitions, that $\cosh 2x = 1 + 2\sinh^2 x$ for every $x$.

(c) Show that $\cosh(\ln x) = \dfrac{x^2+1}{2x}$ for $x > 0$.

(d) Solve the equation $\cosh x = 2$. (Write it in terms of $e^x$.) How many solutions are there, and why was that to be expected?

**Problem 4** *Derivatives of hyperbolic functions.*

(a) $y = \cosh\!\left(x^3\right)$

(b) $y = \ln(1+\cosh x)$

(c) $y = \tanh\!\left(x^2\right)$

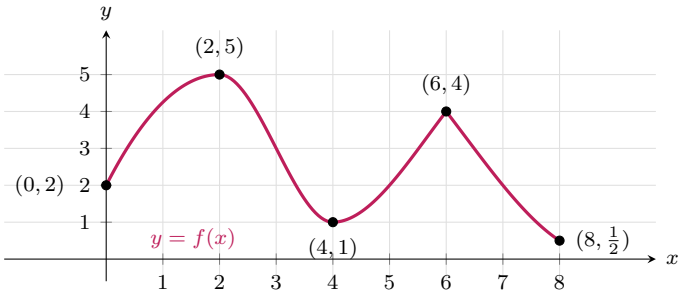
(d) $y = \dfrac{\sinh x}{1+\cosh x}$ (simplify the answer)

(e) Prove two of the three formulas of the lecture's table that were left as homework there:

$$\frac{d}{dx}\operatorname{sech} x = -\operatorname{sech} x\tanh x \qquad \text{and} \qquad \frac{d}{dx}\coth x = -\operatorname{csch}^2 x.$$

**Problem 5** *Extreme values from a graph; the Extreme Value Theorem (homework).*

(a) The graph of a function $f$ with domain $[0, 8]$ is shown. State the absolute maximum and minimum values and where they are taken; all the local maximum and minimum values; and the critical points of $f$ in $(0, 8)$. At which of the five marked points does Fermat's Theorem apply, and what does it say there?




(b) For each function, say whether the Extreme Value Theorem *guarantees* an absolute maximum and an absolute minimum. If it does not, say which hypothesis fails, and whether the extreme values exist anyway.

(i) $f(x) = x^2 - 6x$ on $[0, 7]$;

(ii) $g(x) = \dfrac{1}{x}$ on $(0, 1]$;

(iii) $h(x) = x^2$ for $0 \le x < 2$ and $h(2) = 1$, on $[0, 2]$.

**Problem 6** *Critical points.*

Find all the critical points of each function — the numbers in the domain where $f'$ is 0 *or does not exist.*

(a) $f(x) = 2x^3 - 9x^2 - 24x + 5$

(b) $g(x) = \left(x^2 - 4\right)^{2/3}$

(c) $h(x) = 3\tan^{-1} x - x$

(d) $k(x) = x\, e^{-x^2}$

**Problem 7** *The Closed Interval Method.*

Find the absolute maximum and minimum values of each function on the given interval, and where they are taken.

(a) $f(x) = 2x^3 - 15x^2 + 24x + 3$ on $[0, 3]$

(b) $g(x) = x - 2\sin x$ on $[0, \pi]$

(c) $h(x) = \dfrac{x}{x^2 + 9}$ on $[0, 9]$

(d) The function of 6(b), $g(x) = \left(x^2 - 4\right)^{2/3}$, on $[-1, 3]$.

## Part II · The Mean Value Theorem and the shape of a graph (3.2–3.3) · 50 minutes

**Warm-up II · True or false? (justify with one sentence or one counterexample)**

1. If $f$ is continuous on $[a,b]$ and $f(a)=f(b)$, then there is a number $c$ in $(a,b)$ with $f'(c)=0$.
2. If $f'(x)>0$ for every $x$ in $(a,b)$, then $f$ is increasing on $(a,b)$.
3. If $f'(x)=0$ for every $x$ in the domain of $f$, then $f$ is constant.
4. If the graph of $f$ is concave upward on $(a,b)$, then $f''(x)>0$ for every $x$ in $(a,b)$.
5. If $f''$ is continuous near $c$, $f'(c)=0$ and $f''(c)<0$, then $f$ has a local maximum at $c$.
6. If $f$ is continuous at 2, $f'(x)<0$ for $x<2$ and $f'(x)>0$ for $x>2$, then $f$ has a local minimum at 2.
7. For $f(x)=x^2$ on $[1,3]$, the number $c$ of the Mean Value Theorem is $c=2$.
8. If $f'(x)=g'(x)$ for every $x$, then $f(x)=g(x)$ for every $x$.
9. If $f$ is concave upward on an interval, its graph lies above its tangent lines there.
10. For $f(x)=x^3-9x$ on $[-3,3]$ there is exactly one number $c$ in $(-3,3)$ with $f'(c)=0$.

**Problem 8** *Rolle's Theorem and the Mean Value Theorem: find c.*

In (a)–(c), check that the hypotheses hold, then find *every* number $c$ that the theorem promises.

(a) Rolle's Theorem for $f(x)=\sin 2x$ on $[0,\pi]$.

(b) The Mean Value Theorem for $f(x)=\sqrt{x}$ on $[0,25]$. (Is $f$ differentiable at 0? Does it matter?)

(c) The Mean Value Theorem for $f(x)=\ln x$ on $[1,e]$.

(d) Let $f(x)=|x-1|$ on $[0,4]$. Show that there is *no* number $c$ with $f(4)-f(0)=f'(c)(4-0)$. Why does this not contradict the Mean Value Theorem?

**Problem 9** *What the Mean Value Theorem can prove.*

(a) Show that the equation $x^3+9x-3=0$ has *exactly one* real solution, and find an interval of length $\frac{1}{2}$ that contains it. (At least one: the Intermediate Value Theorem. At most one: Rolle's Theorem.)

(b) Show that, whatever the constant $k$, the equation $x^4+6x+k=0$ has *at most two* real solutions.

(c) A function $f$ is differentiable on $\mathbb{R}$, $f(2)=-1$, and $f'(x)\le 3$ for every $x$. How large can $f(6)$ possibly be? Give a function for which that value is reached.

(d) Prove the identity $\cos^{-1}x+\cos^{-1}(-x)=\pi$ for $-1\le x\le 1$, by differentiating the left side.

**Problem 10** *Increasing or decreasing; the First Derivative Test.*

For each function find the intervals on which it is increasing or decreasing, and its local maximum and minimum values.

(a) $f(x)=2x^3-15x^2+24x+3$ (the function of 7(a), now on the whole line)

(b) $g(x)=x-\sqrt{2}\sin x,\ 0\le x\le 2\pi$

(c) $h(x)=x^4-6x^2+8x$ (factor $h'$ completely; one factor is a square)

(d) $k(x)=x^2-8\ln x$

**Problem 11** *Concavity, inflection points and the Second Derivative Test.*

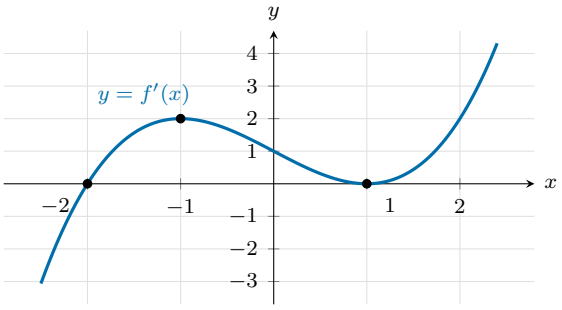
(a) For $h(x) = x^4 - 6x^2 + 8x$ (from 10(c)): find the intervals of concavity and the inflection points; then apply the Second Derivative Test at each critical point — and say what to do where it gives no information.

(b) $g(x) = x\,e^{6x}$: concavity, inflection points, and the local extreme values by the Second Derivative Test.

(c) Let $h(x) = x\tan^{-1} x$. Show that $h''(x) = \dfrac{2}{(1+x^2)^2}$, and deduce the concavity of the graph and the minimum value of $h$.

**Problem 12** *Reading $f'$; sketching from conditions.*

(a) The graph of the *derivative* $f'$ of a function $f$ is shown, for $-2.5 \le x \le 2.4$ (not the graph of $f$). On what intervals is $f$ increasing, and on what intervals decreasing? At which $x$ does $f$ have a local maximum or minimum? On what intervals is the graph of $f$ concave upward or downward, and at which $x$ are its inflection points?




(b) Sketch a possible graph of a function $f$ that satisfies all of the following: (i) $f(1) = -2$ and $f'(1) = 0$; (ii) $f'(x) < 0$ for $x < 1$ and $f'(x) > 0$ for $x > 1$; (iii) $f''(x) > 0$ for $-1 < x < 3$, and $f''(x) < 0$ for $x < -1$ and for $x > 3$; (iv) $\lim\limits_{x\to-\infty} f(x) = 2 = \lim\limits_{x\to\infty} f(x)$. What kind of point is $(1, -2)$, and what happens at $x = -1$ and $x = 3$?

**Challenge** *(for those who finish early, or at home)*

1. Use the Mean Value Theorem to show that $\left|\tan^{-1} a - \tan^{-1} b\right| \le |a - b|$ for all real numbers $a$ and $b$.
2. Let $f(x) = px^2 + qx + r$ with $p \neq 0$. Show that, on *any* interval $[a, b]$, the number $c$ of the Mean Value Theorem is the midpoint $\dfrac{a+b}{2}$ — and give a function for which it is not.
3. The inverse cotangent is defined by $\cot^{-1} x = y \iff \cot y = x$ and $0 < y < \pi$ (for every real $x$). Prove that $\dfrac{d}{dx}\cot^{-1} x = -\dfrac{1}{1+x^2}$, as the lecture proved the formula for $\tan^{-1}$.